The Empirical Bayes Envelope and Regret Minimization in Stochastic Games

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Abstract. In repeated matrix games, classical results establish the existence of *regret minimizing* strategies that secure an average payoff as high as the best-response payoff (the *Bayes Envelope*) relative to the empirical frequencies of the opponent's actions. In this paper we consider the question of regret minimization in the framework of stochastic (Markov) games, under appropriate recurrence properties. Our approach relies on the observed state-action frequencies. The *empirical Bayes envelope* is naturally defined in this space as the best-response to the stationary strategy of the opponent which agrees with the observed frequencies. This envelope is shown to be non-attainable in general, and we therefore put forward its convexification (the CBE) as a feasible goal. The CBE is shown to be attainable, it guarantees a payoff higher than the minimax value when the opponents play is suboptimal, and reduces to the standard Bayes envelope for repeated matrix games. We further show that if the opponent alone affects the state transitions, then the empirical Bayes envelope is itself convex and attainable. Some concrete examples from information and learning theory are shown to fit our framework, as well as a k-th order extension of the Bayes envelope for repeated matrix games.

The analysis is based on a generalization of Blackwell's approachability theory to stochastic games, which holds under the recurrence assumption related to the existence of a single state which is recurrent under all strategies. Extension of the results to games that satisfy communicating-type conditions are outlined.

Keywords: Bayes envelope, Stochastic Games, Regret minimization, Approachability theory.

1 Introduction

Stochastic games are prevalent model for conflict situations in which agents interact in a dynamic environment. With major application areas in economic analysis and operations research, there has been much recent interest in stochastic game models in the context of Artificial Intelligence and machine learning (e.g. [BT95, Pat97, Ras94, Lit94]). Starting from the seminal paper of Shapley [Sha53], there have been an extensive and continuing research on the theory of stochastic games, focusing on the existence of the value, optimal strategies, and their computation; see [FV96] for a recent review.

Game theory has focused on the analysis of conflicts between rational agents, who make strategic decisions based on known individual utilities. It is often the case that a player may not be aware of the utilities of other players, and cannot reason *a-priori* about their strategies. Since the other players need not be completely antagonistic, the "safe" solution of playing a maximin strategy may be too conservative, especially in a repeated or multi-stage game where the actual choices of the players are gradually revealed and may present cues for adaptive play. Obviously, under these circumstances it is desirable to set some feasible goals for our player that rely on the revealed choices of the other players rather than on their intentions.

An elegant formulation of this theme has been presented by the Bayes envelope concept, that was introduced in the context repeated games with average payoff and complete observations. The Bayes envelope is the maximal payoff rate a player could achieve had she known in advance the relative frequencies of the other players. It was originally established by Hannan ([Han57]) that the Bayes envelope may be asymptotically attained in such games. This result was subsequently proved by Blackwell ([Bla56b]) using the theory of approachability for repeated games with vector payoffs. This approach does not rely on any strategic assumptions concerning the opponents, but rather sets out to exploit observed deviations from their respective optimal strategies in order to secure an average payoff which is higher that the maximin value. Strategies that attain the Bayes envelope are often referred to as regret-minimizing (or, more precisely, zero-regret strategies), cf. [FL98].

These classical results rely on a complete observation of the opponent's action in each stage game. Recently, these results have been extended to the case where complete observations of the opponents' actions are not available, but rather some related signals. The papers [ACFS95, FS99] consider the case where the signal includes the reward at each stage, and show that regret minimizing strategies exist in this case as well. This result was further extended in [Rus99] to a general signal structure, where existence of regret minimizing strategies with respect to an appropriately modified Bayes envelope was established. In both cases the Bayes envelope to be ned is defined in the space of the (possibly unobserved) empirical frequencies of the opponent's

The ideas in this paper bare a close resemblance to [MS00b], which considers a game with a general signal structure. The Bayes envelope there was formulated in the reward-signal space. It was shown in [MS00b] that the resulting Bayes like envelope may be not attainable. As a solution concept [MS00b] suggested the convexification of the signal space Bayes envelope. The concepts of [MS00b] are extended in this paper to stochastic games. Namely, defining the Bayes envelope in the reward-observation space and taking its convexification as an attainable solution concept.

Our aim in this paper is to apply the regret-minimization approach to stochastic games. We thus set out to define an attainable goal, in the style of the Bayes envelope, for these games. It is assumed here that complete observations of visited states and the players' actions there are available. We restrict attention to solution concepts that are defined in the space of state-action frequencies (combined with the average reward). Thus, both the convex Bayes envelope defined below and the regret-minimizing strategies that correspond to it depend on these frequencies only. Indeed, in situations of practical interest the state-action frequencies are a natural and readily available summary of the game history.

The Bayes envelope in the state-action-reward space is simply defined as the best reward that can be secured by a player (P1) against the stationary mixed strategy of the opponent (P2) which is compatible with the state-action frequencies. We refer to that set as the *empirical* Bayes envelope, or BE . It turns out, however, that BE is not attainable in general. This was shown even for repeated extensive-form games in [MS00b]. It leads us to consider the convex hull of the BE , the CBE , as the basic solution concept in the state-action space, which is shown to be attainable. The average payoff guaranteed by the CBE is trivially not less than the minimax value of the game; when the opponent plays a sub-optimal stationary strategy the guaranteed payoff is shown to be strictly above the value.

We mention that other notions of Bayes envelope can be given, which parallel those of repeated matrix games by treating the stationary strategies as actions; see Section 6. The realization of such schemes seems to depend on an explicit and repeated experimentation with each possible stationary strategy of P1. The strategies proposed in this paper which attain the CBE do not entail this imposition.

Our basic tool of analysis is Blackwell's approachability theory. Approachability theory has found various uses in game theory and related applications; see, e.g., [FL95, HMC98, Leh98, Spi99] and the recent special issue [GAE99]. The approachability results for dynamic games used in this work are based on the results in [SS93]. In [SS93] a recurrence property of the state dynamics was assumed, namely the existence of a fixed state which is recurrent under all strategies. We adopt this assumption throughout the body of the paper. In Appendix A we replace this property with a weaker one, essentially requiring the recurrence of this state under *some* strategy of P1. We show that both the approachability results and the main results of this paper concerning the

CBE continue to hold under this weaker assumption.

A special case of interest is the single-controller stochastic game, where P2 alone determines the state transitions, is treated in Section 4. These games are closer in nature to repeated matrix game. We show that in this case the BE itself is convex and therefore approachable. We also suggest a simpler scheme which ensures convergence to the BE with improved guaranteed rates. As an immediate application of this framework we examine the concept of *k-th order Bayes envelope* for repeated matrix games, which allows to take advantage of possible correlations in the opponent's actions. Two examples of concrete problems that fit into this framework are analyzed as well, concerning prediction with expert advice ([Vov98] and optimal Markov prediction of finite order.

We should mention here a related body of work which concerns learning processes in matrix games that converge to the (single-stage) equilibrium; a recent survey in [FL98], and further results in [HMC99].

The paper is organized as follows. Section 2 presents the stochastic game model and recalls the basic approachability results required here. In Section 3 we define the empirical Bayes envelope for stochastic games is defined, prove that its convex hull, CBE , is attainable, and establish its basic properties. Section 4 considers the single-controller case. In Section 5 we examine an example of a game in which BE is not attainable by any strategy. In Section 6 we briefly discuss alternative definitions of the Bayes envelope for stochastic game. Some concluding remarks are drawn in Section 7. Appendix A outlines the extension of the main results under a weakened recurrence assumption, based on an appropriate approachability framework. Appendix B provides some calculations that are related to the example of Section 5 and were omitted there for the sake of brevity. Appendix C is dedicated to an example in which the Bayes envelope in the strategy space is not attainable.

2 The General Model

We consider a two-person stochastic game with finite state and action spaces, which is played ad-infinitum. We refer to the players as P1 (the regret-minimizing player) and P2. The game is defined by the five-tuple $(\mathcal{S}, \mathcal{A}, \mathcal{B}, P, r)$, where:

1. $\mathcal{S}$ is the finite set of states of the Markov process, $\mathcal{S} = \{1, \ldots, S\}$.
2. $\mathcal{A}$ is the set of actions of P1 in each state, $\mathcal{A} = \{1, \ldots, A\}$.
3. $\mathcal{B}$ is the set of actions of P2 in each state, $\mathcal{B} = \{1, \ldots, B\}$.
4. P is the state transition kernel. $P : \mathcal{S} \times \mathcal{A} \times \mathcal{B} \times \mathcal{S} \to [0, 1]$ such that $P(s_2|s_1, a_1, b_1)$ is the

probability that the next state is s_2 given that current state is s_1, P1 plays a_1 and P2 plays b_1.

5. r is P1's reward function, $r : \mathcal{S} \times \mathcal{A} \times \mathcal{B} \to \mathbb{R}$. $r(s,a,b)$ is the reward obtained when P1 plays a, P2 plays b and the current state is s.

At each time epoch n, both players observe the current state s_n, and then P1 and P2 choose actions a_n and b_n, respectively. As a result P1 receives a reward of $r_n = r(s_n, a_n, b_n)$ and the next state is determined according to the transition probability $P(\cdot|s_n, a_n, b_n)$. A strategy $\sigma_1 \in \Sigma_1$ for P1 is a mapping from all possible histories to the mixed action $\Delta^{\mathcal{A}}$, where $\Delta^{\mathcal{A}}$ is the set of all probability measures over $\mathcal{A}$. Similarly, a strategy $\sigma_2 \in \Sigma_2$ for P2 is a mapping from all possible histories to the mixed action $\Delta^{\mathcal{B}}$. For simplicity and without loss of generality we restrict attention to *behavioral* strategies where the mixing mechanisms at different stages are independent. A strategy of either player is called *stationary* if the mixed action in time n depends only on the state s_n. The set of stationary strategies of P1 (resp. P2) is denoted by F (resp. G), and the set of stationary deterministic strategies of P1 is denoted by F_D. Let $P^s_{\sigma_1,\sigma_2}$ denote the probability measure induced on the sequence of states and actions by the strategy pair σ_1 and σ_2 and initial state s, and let $E^s_{\sigma_1,\sigma_2}$ denote the corresponding expectation operator. We are primarily interested in the average reward:

$$\hat{r}_n = \frac{1}{n}\sum_{l=1}^{n} r_l \,. \tag{2.1}$$

We shall also consider a related vector-valued stochastic game that will be specified in the sequel. We denote the corresponding reward vector at time n by $m_n = m(s_n, a_n, b_n) \in \mathbb{R}^k$. This reward may be stochastic, in which case $m(s_n, a_n, b_n)$ denotes its mean and the reward is assumed to have a finite second moment. In a similar manner to (2.1) the average vector valued reward is defined as $\hat{m}_n = \frac{1}{n}\sum_{l=1}^{n} m_l$. Given the vector-valued stochastic game and a vector $u \in \mathbb{R}^k$, we define the projected game $\Gamma_s(u)$ as the zero-sum stochastic game with (scalar) reward given by $r_n = m_n \cdot u$ and initial state s. $\Gamma_s(u)$ has a value $\mathrm{v}\Gamma_s(u)$ if

$$\begin{aligned} \mathrm{v}\Gamma_s(u) &= \sup_{\sigma_1}\inf_{\sigma_2}\liminf_{n\to\infty} E^s_{\sigma_1\sigma_2}(\hat{m}_n \cdot u) \\ &= \inf_{\sigma_2}\sup_{\sigma_1}\limsup_{n\to\infty} E^s_{\sigma_1\sigma_2}(\hat{m}_n \cdot u)\,. \end{aligned} \tag{2.2}$$

As is well known (e.g. [FV96, Section 5.2]) this value exists for any finite stochastic game. Our basic assumption regarding the state transition structure of the game is the following:

Assumption 1 *There exists a state $0 \in \mathcal{S}$ that is recurrent under any pair f, g of stationary non-randomized strategies.*

This assumption is usually straightforward to verify by inspecting the transition structure. Recall [PS77] that under this assumption, the value of the game from all starting states is the same. We therefore denote the game in u direction by $\Gamma(u)$ and omit the s sub-index. Denote by τ the first passage time to state 0, that is

$$\tau \triangleq \inf_{n>0}\{s_n = 0\}\,.$$

A strategy $\sigma_1 \in \Sigma_1$ is called *stable* with respect to an initial state s if $\exists M < \infty$ such that:

$$E^s_{\sigma_1\sigma_2}(\tau^2) < M\,, \tag{2.3}$$

for any strategy σ_2 of P2. A set of strategies $\Pi \subseteq \Sigma_1$ is called *uniformly stable* with respect to state s if every strategy in Π is stable with the same constant M in (2.3). Since our model is finite and Assumption 1 holds, Lemma 2.1 in [SS93] implies that every stationary strategy for either player is uniformly stable.

We now recall the following definition that was first introduced in [Bla56a], and later extended to stochastic game in [SS93]. Let $d(\cdot,\cdot)$ denote the Euclidean norm in $\mathbb{R}^k$. Recall that given a set of probability measures $\{P_\alpha\}_{\alpha\in\Upsilon}$ defined on a common measurable space, and a series X_n of random variables on that space, we say that $X_n \to 0$ almost surely at a uniform rate over Υ if $\forall\epsilon > 0$

$$\lim_{N\to\infty}\sup_{\alpha\in\Upsilon} P_\alpha\left(\sup_{n\geq N}|X_n| > \epsilon\right) = 0\,.$$

Definition 2.1 *A set $B \subseteq \mathbb{R}^k$ is* approachable *from a state s by P1 if there exists a B-*approaching *strategy σ_1^* such that*

$$d(\hat{m}_n, B) \to 0 \quad P^s_{\sigma_1^*,\sigma_2}\text{-a.s. for every } \sigma_2 \in \Sigma_2$$

at a uniform rate over $\sigma_2 \in \Sigma_2$ (i.e. over $\{P_{\sigma_1^*\sigma_2}\}_{\sigma_2\in\Sigma_2}$). B is excludable *from s by P2 if there exists a B-*excluding *strategy $\sigma_2^* \in \Sigma_2$ such that for some $\delta > 0$*

$$\liminf_{n\to\infty} d(\hat{m}_n, B) > \delta \quad P^s_{\sigma_1,\sigma_2^*}\text{-a.s. for every } \sigma_1 \in \Sigma_1$$

at a uniform rate over $\sigma_1 \in \Sigma_1$ (i.e. over $\{P^s_{\sigma_1,\sigma_2^*}\}_{\sigma_1\in\Sigma_1}$).

To state the main approachability results, let

$$\phi(\sigma_1,\sigma_2) = \frac{E^0_{\sigma_1,\sigma_2}(\sum_{n=0}^{\tau-1} m_n)}{E^0_{\sigma_1,\sigma_2}(\tau)} \tag{2.4}$$

denote the average payoff per cycle from state 0 and back. Note that ϕ is well defined if either σ_1 or σ_2 is stable, and in particular when Assumption 1 holds. Let B be a closed set in $\mathbb{R}^k$, for any $x \notin B$ we denote by C_x a closest point in B to x. Let H_x be the hyperplane through C_x that is perpendicular to the line $C_x - x$ and u_x be the unit vector in direction $C_x - x$. The following

result is Blackwell's theorem for stochastic games, and is a reduction of Theorem 3.1 of [SS93] to the case of a finite state space, since for that case all stationary strategies are uniformly stable under Assumption 1.

Theorem 2.1 *[SS93, Theorem 3.1] Let Assumption 1 hold. Assume that $\forall x \notin B$ there exists a stable strategy $\sigma_1(x) \in \Sigma_1$ such that:*

$$(\phi(\sigma_1(x), \sigma_2) - C_x) \cdot u_x \geq 0 \quad \forall \sigma_2 \in \Sigma_2 \,. \tag{2.5}$$

Then B is approachable by P1 from any state s.

In [SS93] a B–approaching strategy is presented. The main idea there is to switch strategies at every return to the reference state in such a way that the strategy is optimal for the projected game in the direction u_x. We refer the reader to that paper for a detailed description.

Convex sets may be categorized as either approachable or excludable ([Bla56a]). For every pair of stationary strategies f of P1 and g of P2, the long-run average expected payoff $M(f,g) \stackrel{\Delta}{=} \lim_{n\to\infty} E_{f,g}\hat{m}_n$ is well defined under Assumption 1. We also define the following set for any $g \in G$: $M(F,g) \stackrel{\Delta}{=} \mathrm{co}(\{M(f,g), f \in F\}) \subseteq \mathbb{R}^k$. The next theorem characterizes convex approachable sets.

Theorem 2.2 *[SS93, Theorem 3.4] Let Assumption 1 hold. Let B be a closed convex set in $\mathbb{R}^k$.*

(i) B is approachable if and only if either one of the following equivalent conditions are satisfied:

a. $M(F,g) \cap B \neq \emptyset$ for every stationary strategy $g \in G$.

b. $\mathrm{v}\Gamma(u) \geq \min_{y\in B}(y \cdot u)$ for every $u \in U(B)$. Here $U(B) \subset \mathbb{R}^k$ is the set of all unit vectors u in $\mathbb{R}^k$ such that $\exists x \notin B$ for which $C_x - x$ is in the u direction.

(ii) If B is not approachable then it is excludable by P2 with a stationary strategy.

Remark 1: The basic prerequisite for the approachability results above is the independence of the minimax value and the initial state. Assumption 1, besides implying that property, also identifies a reference state at which P1 may switch between stationary strategies. In the Appendix we relax this assumption, and require merely that the reference state may be reached from any other state by *some* strategy of P1 (a requirement which is akin to a communicating chain structure). The basic approachability results still hold, although the approaching strategy is complicated by the need to increase the interval between subsequent stationary strategy switches in an unbounded manner.

Remark 2: Another issue that is related to stability of the reference state is the rate of convergence. Under Assumption 1 one can obtain a convergence rate that satisfies:

$$P\left\{\sup_{n\geq N} d(\hat{m}_n, B) > \epsilon\right\} \leq \frac{Q}{\epsilon^2 N}$$
$$E\left(d(\hat{m}_n, B)^2\right) < \frac{Q'}{n} \tag{2.6}$$

where Q, Q' depend only on the parameters of the stochastic game, and in particular on the stability constant M. This may be derived in a similar manner to [MSZ94, page 126]. Furthermore, it may be shown that $d(\hat{m}_n, B)$ behaves almost surely like $\Theta(n^{-1/3})$ (we say that a series f(n) behaves like $\Theta(n^k)$ if $\liminf_{n\to\infty} \log(f(n))/log(n) \leq k$), see e.g. [MSZ94, page 356].

3 The Empirical Bayes Envelope for Stochastic Games

In this section we define the empirical Bayes envelope for stochastic games and prove our main result regarding the attainability of its lower convex hull. This definition of the empirical Bayes envelope coincides with the standard definition of the Bayes envelope in the special case of repeated matrix games.

We denote by $\pi_n(s,b)$ the empirical frequency of P2's play of action b in state s by time n, namely

$$\pi_n(s,b) = \frac{1}{n}\sum_{l=1}^{n} 1\{s_l = s, b_l = b\},$$

and $\pi_n \in \Delta^{SB}$ is the state-action frequencies vector. Let Π_2 denote the realizable set of state-action frequencies, namely the closed convex hull of the set of limit points of $\{\pi_n\}$ under any pair of strategies. An *envelope* is a set of the form $B = \{(r,\pi) \in \mathbb{R} \times \Pi_2 : r \geq h(\pi)\}$ where h is the *envelope function.* The following definition specifies a required payoff performance in the state-action-reward space.

Definition 3.1 *Let the initial state $s \in S$ be fixed. The envelope $B = \{(r,\pi) : r \geq h(\pi)\} \subseteq \mathbb{R}\times\Pi_2$ is* attainable *from s by P1 if there exists a B-*attaining *strategy σ_1^* such that*

$$\liminf_{n\to\infty}(\hat{r}_n - h(\pi_n)) \geq 0 \quad \text{P-a.s. for every } \sigma_2 \in \Sigma_2$$

at a uniform rate over σ_2 (i.e. over $\{P^s_{\sigma_1^*,\sigma_2}\}_{\sigma_2\in\Sigma_2}$).

We now turn to define the empirical Bayes envelope, in terms of the average reward and the empirical state-action frequencies. If P2's strategy was stationary, it could be naturally estimated from the relative frequencies π by:

$$g(\pi)(b|s) = \frac{\pi(s,b)}{\pi(s)}, \tag{3.1}$$

where $\pi(s) \triangleq \sum_{b=1}^{B} \pi(s,b)$ and $g(b|s)$ is the probability of picking action b in state s. Since $\pi(s)$ may be 0, we consider the set of all possible strategies that are consistent with π, that is:

$$G(\pi) = \{g \in G : \pi(s)g(b|s) = \pi(s,b) \ \forall (s,b) \in \mathcal{S} \times \mathcal{B}\}. \tag{3.2}$$

$G(\pi)$ is a singleton when every state has been visited, and in any case is convex and closed. We can now formulate the Bayes envelope in the reward-state-action space:

$$BE \triangleq \{(r,\pi) \in \mathbb{R} \times \Pi_2 : r \geq r^*(\pi) \triangleq \max_{f \in F} \min_{g \in G(\pi)} r(f,g)\} \tag{3.3}$$

where $r(f,g)$ is the expected average reward of the two stationary strategies f and g. Note that $r(f,g)$ (and similarly $m(f,g)$ below) is well defined under Assumption 1, e.g. [Put94, Proposition 8.1.1]. $r^*(\pi)$ can be seen as the best-response payoff of P1 to the strategies $G(\pi)$. It is obvious that $r^*(\pi)$ is never below the minimax value of the game, since the minimum in (3.3) is over a subset of G. It is strictly above the value if $G(\pi)$ does not contain an optimal strategy for P2.

Unfortunately the envelope BE just defined need not be attainable; see the example in Section 5. We shall therefore consider a larger set as an attainable goal.

The *Convex Bayes Envelope*, $CBE \triangleq \text{co}(EBE)$, defined as convex hull of BE . Noting that the set BE is the epigraph of the function r^*, CBE can be written as $CBE = \{(r,\pi) : r \geq r^c(\pi)\}$ where $r^c(\pi)$ is the lower convex hull of r^*. It will be shown below that CBE is indeed attainable. We start with some continuity properties of the envelope functions.

Proposition 3.1 *Under Assumption 1 the following hold:*

(i) $r^* : \Pi_2 \to \mathbb{R}$ *is continuous on all points of* $\Pi_2 \subset \Delta^{SB}$ *that are in* $ri(\Delta^{SB})$ *(the relative interior of* Δ^{SB}*), and lower semi continuous elsewhere.*

(ii) r^c *is continuous on* Π_2.

(iii) For all π, $r^c(\pi) \geq \text{v}$, *where* v *is the value of the zero-sum game.*

Proof:

(i) For $\pi \in \text{ri}(\Delta^{SB})$ we have $\pi(s) > 0$ for all s, so that the set $G(\pi)$ is a singleton, $g(\pi)$. The function $r^*(\pi)$ of (3.3) can be written as:

$$r^*(\pi) = \max_{f \in F} r(f, g(\pi)).$$

It is well known that for a Markov Decision Process with average reward there is a maximizing strategy which is stationary and deterministic (e.g. [FV96] Corollary 2.4.5). We

can therefore assume that the maximum is over the finite set of deterministic stationary strategies of P1 ($f \in F_D$). In order to prove continuity of r^* on the relative interior of Δ^{SB} it suffices to show that for each $f \in F_D$ the following functions is continuous there:

$$h_f(\pi) \stackrel{\Delta}{=} r(f, g(\pi)) .$$

The average reward of two stationary strategies f and g is

$$r(f,g) = \sum_s \sum_a \sum_b \pi_{fg}(s) f(a|s) g(b|s) r(s,a,b) ,$$

where $f(a|s)$ is the probability that P1 plays action a at state s, $g(b|s)$ is the probability that P2 plays action b at state s, and $\pi_{fg}(s)$ is invariant probability measures of the Markov process that is induced when P1 plays f and P2 plays g. Obviously $g(\pi)$ is continuous in $\mathrm{ri}(\Delta^{SB})$, as may be seen from its definition. Moreover, under Assumption 1, π_{fg} is unique and continuous in g (see e.g. [Put94, Proposition 8.4.6]). We therefore conclude form the last equation that $r^*(\pi)$ is continuous in $\mathrm{ri}(\Delta^{SB})$.

At the boundaries the continuity may not hold, as $G(\pi)$ may not be a singleton anymore. To establish lower semi continuity, fix $\pi \in \Pi_2$, and consider a sequence $\pi_n \to \pi$. It is required to show that $\liminf_{n\to\infty} r^*(\pi_n) \geq r^*(\pi)$. For that purpose it is enough to show that this inequality holds for any subsequence π_{n_k} so that $r^*(\pi_{n_k})$ converges. Fixing such a subsequence, and denoting it by π_m (with some abuse of notation), let $f_m \in F$ and $g_m \in G(\pi_m)$ be a pair of strategies that jointly achieve the maximum and minimum, respectively, in (3.3). Thus $r^*(\pi_m) = r(f_m, g_m)$. Take a further subsequence π_{m_l} so that f_{m_l} and g_{m_l} converge to some f^* and g^*, respectively (such subsequence exist since P2, F and G are compact). Then

$$\lim_{m\to\infty} r^*(\pi_m) = \lim_{l\to\infty} r^*(\pi_{m_l}) = \lim_{l\to\infty} r(f_{m_l}, g_{m_l}) = r(f^*, g^*)$$

where the last inequality follows by continuity of r^* in f and g as noted above. Now from $g_{m_l} \in G(\pi_{m_l})$ and the definition of $G(\pi)$ it follows that $g^* \in G(\pi)$. Moreover, since f_{m_l} maximizes $r(f, g_{m_l})$, then by continuity f^* maximizes $r(f, g^*)$. Thus,

$$r(f^*, g^*) = \max_f r(f, g^*) \geq \max_f \min_{g\in G(\pi)} r(f,g) = r^*(\pi) .$$

Together with the previous equation this establishes the required inequality.

(ii) Since Π_2 is a closed convex set and r^c a finite convex function on it, it follows that r^c is continuous on $\mathrm{ri}(\Pi_2)$, (cf. [Roc70, Theorem 10.2]). Continuity at the boundaries follows from r^* being lower semi continuous and by [Roc70, Theorem 10.3].

(iii) This follows by recalling that $r^*(\pi) \geq \mathrm{v}$. **Q.E.D.**

In order to apply approachability arguments in the sequel, we introduce the following vector-valued payoff for this game. Define the $1 + SB$ dimensional reward vector $m = (m_r, m_\pi) \in \mathbb{R} \times \Delta^{BS}$, where m_π is indexed by the state-action pairs (s, b). Given that the state is s, P2 played action b and the observed reward was r, the payoff m is given by

$$m = (r, e_{sb}) \tag{3.4}$$

where e_{sb} is a unit vector with 1 at the entry corresponding to (s, b). Thus, for the average payoff vector $\hat{m}_n = \frac{1}{n}\sum_{l=1}^{n} m_n$ we obtain $\hat{m}_n = (\hat{r}_n, \pi_n)$, namely the average scalar reward and the vector of state-action frequencies. For a pair of stationary strategies f of P1 and g of P2 we denote the average vector reward by $m(f, g)$.

Attainablilty of an envelope $E = \{(r, \pi) : r > h(\pi)\}$ is closely related to approachability of that set with the above vector payoff. Approachability requires that the Euclidean distance of $\hat{m}_m = (\hat{r}_n, \pi_n)$ from E converges to zero, while attainability requires the reward coordinate difference $\hat{r}_n - h(\pi_n)$ to become non-negative. When h is continuous, the two notions are obviously equivalent (while otherwise it is only true that attainability implies approachability).

Theorem 3.2 *Suppose Assumption 1 holds. Then the envelope CBE is attainable by P1.*

Proof: From Proposition 3.1 we know that r^c is continuous on Π_2. As just noted, it is therefore sufficient to show that CBE is approachable by P1 with the payoff vector m. Since CBE is convex it suffices by Theorem 2.2 to prove that $\forall g \in G$, $M(F, g) \cap CBE \neq \emptyset$. Fix $g^* \in G$, in the original scalar game P1 has a best-response stationary (deterministic) strategy $f^* \in F$ against g^*. Consider the state-action frequency π that is the invariant measure corresponding to f^* and g^*. We now show that the point $m^* \stackrel{\Delta}{=} (r(f^*, g^*), \pi) \in M(F, g^*)$ is in BE and therefore also in CBE . By (3.2) and the definition of π, $g^* \in G(\pi)$. The reward $r^*(\pi)$ thus satisfies

$$r^*(\pi) = \max_{f \in F} \min_{g \in G(\pi)} r(f, g) \leq \max_{f \in F} r(f, g^*) = r(f^*, g^*) \tag{3.5}$$

so that $m^* \in BE$ by definition of the latter. **Q.E.D.**

We now consider the average reward guaranteed to P1 by attaining CBE . Since $r^*(\pi) \geq \mathrm{v}$ (where v is the min-max value of the game), the same it true for r^c (the lower convex hull of r^*). Thus the guaranteed average reward is never lower than the value. We would like to show that when P2's observed play (as evidenced by the relative frequencies of his actions at the states visited) is suboptimal, then the guaranteed reward is larger than v. We shall establish that for the class of *irreducible* stochastic games. Recall that a stochastic game is irreducible if all states are recurrent under every pair of (deterministic) stationary policies. It follows that $\pi(s) > 0$ for every s and $\pi \in \Pi_2$. In particular, each feasible $\pi \in \Pi_2$ induces a single stationary strategy, denoted $g(\pi)$, according to (3.2).

Proposition 3.3 *Suppose the game is irreducible. Let the set of stationary min-max optimal strategies for P2 be denoted by $G^* \subseteq G$. Then*

(i) $r^c(\pi) > \mathrm{v}$ *for any* $\pi \in \Pi_2$ *such that* $g(\pi) \notin G^*$.

(ii) Furthermore, $\tilde{v}(g) \stackrel{\Delta}{=} \inf_{\pi: g \in G(\pi)} r^c(\pi) > \mathrm{v}$ *for* $g \notin G^*$.

Proof: The proof relies on the structure of the set G^* of maximin optimal policies for P2 under the irreducibility assumption. We first show that G^* is a Cartesian product of optimal (convex) sets of actions at each state. ¿From [Pat97, Proposition 5.1] we know that there is a unique (up to an additive constant) $w^* \in \mathbb{R}^S$ such that $\forall s \in \mathcal{S}$

$$\min_{g \in G} \max_{f \in F} \left(\sum_a \sum_b f(a|s) g(b|s) r(s,a,b) + \sum_a \sum_b \sum_{s'} P(s'|s,a,b) w^*(s') \right) = w^*(s) + \mathrm{v}\,. \tag{3.6}$$

where v is the value of the game. We now apply [FV96, Lemma 5.3.1] twice. First, note that the assumptions of Lemma 5.3.1 holds since the game is irreducible. We claim that $g \in G^*$ if and only if for every s the mixed action $\{g(s,\cdot)\}$ is in the set $G^*(s)$ of optimal strategies in the matrix game with the following reward for actions a, b:

$$r_{ab}(s) = \sum_a \sum_b r(s,a,b) + \sum_{s'=1} p(s'|s,a,b) w^*(s')\,. \tag{3.7}$$

Suppose g is optimal in every such game. Then from part (iii) of Lemma 5.3.1, for every $f \in F$ we have that $r(f,g) \leq \mathrm{v}$, so that g is optimal. Suppose that g is not optimal for a matrix game that is induced by some state s'. There is a strategy for P1, f^* that promises a higher reward than the value of the game defined in (3.7) for P1. Using the other direction of part (iii) of Lemma 5.3.1 (for P1) we have that $r(f^*,g) \geq \mathrm{v}$, but from part (iv) of Lemma 5.3.1 we get that $r(f,g) > \mathrm{v}$ since equality does not hold for all states. As a result the set G^* is a Cartesian product of the optimal strategy sets at each state, namely $G^* = \bigotimes_s G^*(s)$. Furthermore each set $G^*(s)$ is convex, which follows from the well known fact that for a zero sum matrix game the set of optimal strategies is convex.

Suppose $g \notin G^*$ and let $(r,\pi) \in CBE$ such that $g(\pi) = g$. Since $g \notin G^*$, then $r^*(\pi) > \mathrm{v}$. Since for any point in BE the reward is not less than the value we have that if $(\mathrm{v},\pi) \in CBE$ then there exist at most k ($k \leq SB+1$) points in BE , $\{(\mathrm{v},\pi_i)\}_{i=1}^k$ such that $g(\pi_i) \in G^*$ and for some $0 < \alpha_i < 1$ we have that $\pi = \sum_{i=1}^k \alpha \pi_i$. It can be verified by trivial algebra that if $\pi = \sum_{i=1}^k \alpha_i \pi_i$ then $\forall s$ such that $\pi(s) > 0$ we have that there exists β_i such that $0 \leq \beta_i \leq 1$, $\sum_{i=1}^k \beta_i = 1$ and $\forall b$:

$$g(\pi)(b|s) = \sum_{i=1} \beta_i g(\pi_i)(b|s)\,,$$

specifically $\beta_i = \frac{\alpha_i \pi_i(s)}{\sum_{i=1}^{k} \alpha_i \pi_i(s)}$. The result follows since G^* is a Cartesian product and if $g(\pi_i)$ belong to G^* so does any strategy in the SB dimensional box between them. **Q.E.D.**

A completely different proof to the above uses the Stochastic Shortest Path formulation. It can be shown that the associated stochastic shortest path game with reward $r(s,a,b) - \mathrm{v}$ has the same optimal strategy as the average cost game. We refer the reader to [Ber95, page 83], for average cost reward (not in games) and the proof or [Pat97, Proposition 5.1] for the game relation. The above proof fails to hold when there are states that are transient for certain strategies. Indeed, if $g(\pi)$ is not uniquely determined then $\tilde{v}(g)$ may be of little interest. The following example satisfies that $\tilde{v}(g) = \mathrm{v}$ for every strategy g, even though the best response reward to some of the strategy is larger than v. Let the states space be defined by $\mathcal{S} = \{x, y, z\}$; the actions of P1 at state x are $\mathcal{A} = \{1, 2\}$; these actions determine the transition so that $p(s' = y | s = x, a = 1) = 1$ and $p(s' = z | s = x, a = 2) = 1$. The transition probability from states y, z satisfy $p(s' = x | s = y) = 1$ and $p(s' = x | s = z) = 1$. The reward for states x and y is constant, that is: $r(x, \cdot, \cdot) = r(y, \cdot, \cdot) = 3/4$. P1 and P2 actions in state z form a "matching penny" game. P2 effects the reward of state z alone so that $r(z, 1, 1) = r(z, 2, 2) = 1$ and $r(z, 2, 1) = r(z, 1, 2) = 0$. The state action frequency are: $(\pi_x, \pi_y, \pi_{z1}, \pi_{z2})$. It can be shown that the point $\pi_0 = (1/2, 1/2, 0, 0)$ is a point in BE . However, $r^*(\pi_0) = 3/4 = \mathrm{v}$, and moreover, $G(\pi_0) = G$ so that for the above example $\tilde{v}(g) = \mathrm{v}$ for all g. Note, however, that for some $\pi \in \Pi_2$ a higher reward is guaranteed.

In proving that CBE is attainable (Theorem 3.2) we used the fact that a convex set is approachable if and only if it intersects every set of the form $M(F, g)$. We can use that observation to define smaller sets than CBE which still satisfy this property and are thus approachable.

Proposition 3.4 *Let BE_0 be an envelope such that for every $g \in G$ there is $f(g) \in F$ which is optimal against g, i.e. $r(f(g), g) = \sup_{f \in F} r(f, g)$ and $m(f(g), g) \in BE_0$. Then the convex hull $CBE_0 \triangleq \mathrm{co}(BE_0)$ is approachable and the resulting envelope is attainable.*

Proof: The proof follows in the exact same manner as Theorem 3.2. **Q.E.D.**

Obviously, we can choose $CBE_0 = \mathrm{co}(\{m(f(g), g)\}_{g \in G})$, and in this case $CBE_0 \subseteq CBE$. Furthermore, such a CBE_0 can be constructed to be a convex polyhedron by a proper choice of $f(g)$. Indeed, since for every $g \in G$ there is a maximizing deterministic stationary strategy it suffices to consider in CBE_0 the convex hull of $m(f, g)$ for which f is stationary and deterministic. Recall that the reward $m(f, g)$ for a fixed f is a polyhedron in the reward-state-action space ([Put94, Theorem 8.9.4]), so that CBE_0 can be chosen as the convex hull of a finite number of polyhedras. This implies that by choosing deterministic $f(g)$, the envelope CBE_0 may be characterized as a convex hull of a finite number of points. This raises the possibility of using CBE_0 as a concrete object in finding optimal strategies in applications.

Since $CBE_0 \subseteq CBE$ the payoff guaranteed by CBE_0 is at least that guaranteed by CBE (at each π), and similarly for the associated payoff guarantee ($\tilde{v}(g)$) defined in Proposition 3.3. In fact it may be strictly larger as the following example demonstrates. The example is a modification of the example that follows Proposition 3.3, replacing the reward in states x and y to $1/2$. We have that the CBE satisfies that $\tilde{v}(g) = \mathrm{v} = 1/2$ for every $g \in G$. After this modification of the game we can now define CBE_0 as the envelope which is induced by $(1/2, 0, \frac{\epsilon}{2}, \frac{1-\epsilon}{2})$, $0 \leq \epsilon \leq 1$. The attainable reward for every such element is the Bayes reward since this is actually a matrix game and the CBE_0 is simply given by the envelope function

$$r_0^c(1/2, 0, \frac{\epsilon}{2}, \frac{1-\epsilon}{2}) = 1/4 + 1/2 \max\{\epsilon, 1-\epsilon\} .$$

It can be shown that the corresponding envelope $\tilde{v}(g)$ envelope (defined as in part (ii) of Proposition 3.3) satisfies:

$$\tilde{v}_0(g) \stackrel{\Delta}{=} \inf_{\pi : g \in G(\pi)} r_0^c(\pi) = r^*(g) .$$

Given that CBE is not a minimal approachable set, a question that remains open concerns "optimal" reductions of CBE . Relevant notions of optimality are the Pareto efficiency of the guaranteed reward – either in the frequency space, compared with $r^*(\pi)$ or in the strategy space, compared with $\tilde{v}(g)$. This topic is under current investigation.

4 Single Controller Games

In this section we consider the special case in which P1 does not affect the state transitions, that is $P(s'|s,a,b) = P(s'|s,b)$. The resulting model can be considered as a sequence of matrix games where the next game to be played is determined only by P2's action. These models are often called single controller games ([FV96]) as only one of the players controls the state transitions. As it turns out, from the regret minimization perspective P1 need not concern himself here with the state transitions as the stochastic game can be partitioned into a sequence of interleaved repeated matrix games. Thus, performing optimally in each matrix game suffices for performing optimally in the overall game. Proposition 4.1 states that BE is convex and therefore attainable, in Subsection 4.1 we provide a general result which applies without any recurrence assumption and provides a better convergence rate. As an immediate application of these results, we examine in Subsection 4.2 the k-th order Bayes envelope for repeated matrix games, which improves on the standard definition of the Bayes envelope by taking into account possible correlations in the opponent's actions. This envelope is shown to be attainable by an appropriate embedding of the repeated matrix game in a single controller stochastic game. In Subsection 4.3 we present two additional examples for applications of the single controller game framework to specific problems.

We begin with a characterization of the set BE for games in which P2 controls the state transitions.

Proposition 4.1 *Let Assumption 1 hold. Suppose that P1 does not effect the state transitions, i.e.* $P(s'|s,a,b) = P(s'|s,b)$. *Then* BE *is convex.*

Proof: Under the assumption that $P(s'|s,a,b) = P(s'|s,b)$, the stochastic game dynamics reduces to a Markov Decision process with a single decision maker. Let $\Pi_0 \subset \Delta^{SB}$ be the set of possible state-action frequencies for P2. It is well known (e.g. [Put94, Theorem 8.9.4]) that Π_0 is a convex set. Since only P2 effects the transition dynamics, then for a given π for every $g \in G(\pi), f \in F$, the state action frequencies $\pi(f,g) = \pi$. Furthermore, the reward $r(f,g)$ can be written as the sum:

$$r(f,g) = \sum_s \sum_b \sum_a \pi(s,b) f(a|s) r(s,a,b)\,, \tag{4.1}$$

where $f(a|s)$ is the probability of playing action a at state s for the stationary strategy f. It follows that the optimal reward $r^*(\pi)$ reduces to:

$$r^*(\pi) = \max_{f \in F} \sum_s \sum_b \sum_a \pi(s,b) f(a|s) r(s,a,b) \tag{4.2}$$

This implies that BE is in fact a convex set since it is the maximum of linear functions. **Q.E.D.**

Since BE is convex we can use the same arguments as in Theorem 3.2 to show that BE is approachable and therefore attainable. We therefore have the following corollary:

Corollary 4.2 *For single controller games controlled by P2,* BE *is attainable.*

The above result was obtained using the general approachability framework, so that the guaranteed rate of convergence to BE is $\Theta(n^{-1/3})$. By using specific analysis, Theorem 4.3 generalizes this result so it does not depend on Assumption 1 and provides an improved rate of convergence.

4.1 Generalized Results With Improved Convergence Rate

In this sub-section we show the empirical Bayes envelope for every single controller stochastic game is, attainable without resorting to Assumption 1. We offer a simple algorithm that ensures that BE is attained. This method takes full advantage of the fact that the single controller game can be effectively partitioned into a set of repeated matrix games, each corresponding to a single state. Let us re-define $r^*(\pi)$ (and BE) so that it will be well defined without any assumptions on the game dynamics:

$$r^*(\pi) \stackrel{\Delta}{=} \max_{f \in F} \sum_s \sum_b \sum_a \pi(s,b) f(a|s) r(s,a,b)\,. \tag{4.3}$$

If Assumption 1 holds then obviously the original definition of r^* in equation (4.2) is the same as (4.3) for every feasible state action frequency vector π. The set BE is now defined as in (3.3) as the envelope of r^*. We can now state a general result for single controller games.

Theorem 4.3 *Assume that P1 does not effect the state transitions, i.e.* $P(s'|s,a,b) = P(s'|s,b)$. *Then:*

i. BE *is attainable.*

ii. The convergence rate satisfies $\hat{r}_t - r^*(\pi_t) > -Ct^{-1/2}$ *P-a.s., where* C *is a constant that depends on the game.*

Proof: BE is attained if the average reward $\hat{r}_t$ and the state action frequencies π_t satisfy that

$$\liminf_{n\to\infty}(\hat{r}_t - r^*(\pi_t)) \geq 0 \text{ P-a.s.}. \tag{4.4}$$

Note that since this is a single controller game then $\pi(f,g) = \pi(g)$, that is the state action frequencies depend only on P2's strategy. Starting from equation (4.3), and using the convexity of the max operator we have that:

$$\begin{aligned} \hat{r}_t - r^*(\pi_t) &= \hat{r}_t - \sum_s \max_{f\in F} \sum_b \sum_a \pi_t(s,b) f(a|s) r(s,a,b) \\ &- \hat{r}_t - \sum_s \max_a \sum_b \pi_t(s,b) f(a|s) r(s,a,b)\,, \end{aligned} \tag{4.5}$$

where $\pi_t(s,b)$ is the relative frequency of state s and action b as measured at time t. The last equality is justified since f affects the inner sum only through the actions at state s. But $\hat{r}_t = \sum_s \sum_b \sum_a \pi_t(s,b) f(a|s) r(s,a,b)$ so

$$\hat{r}_t - r^*(\pi_t) = \sum_s \pi_t(s) \left(\sum_b \sum_a g_t(b|s) f(a|s) r(s,a,b) - \max_a \sum_b g_t(b|s) f(a|s) r(s,a,b) \right), \tag{4.6}$$

where $g_t(b|s) \triangleq \frac{\pi_t(s,b)}{\pi_t(s)}$ and $\pi_t(s) = \sum_b \pi_t(s,b)$. Suppose P1's strategy is to play a regret minimizing strategy for every state s separately for the game in which P1's actions are $a \in \mathcal{A}$, P2's actions are $b \in \mathcal{B}$ and the expected reward is $r(s,a,b)$. In that case, (4.6) becomes the sum of elements that tend to zero as $t \to \infty$ (if the state is visited often enough). We have that

$$\hat{r}_t - r^*(\pi_t) \geq \sum_s \pi_t(s) R_t(s)\,, \tag{4.7}$$

where $R_t(s)$ is the regret of the s game by time t:

$$R_t(s) \triangleq \sum_b \sum_a g_t(b|s) f(a|s) r(s,a,b) - \max_a \sum_b g_t(b|s) f(a|s) r(s,a,b)\,.$$

Now, P1's strategy ensures that $R_t(s) \to 0$ almost surely for every state s that is visited infinitely often, while $\pi_t(s) \to 0$ for the other states. It easily follows from (4.7) that (4.4) holds.

The second part follows by using a regret minimizing algorithm that satisfies a regret bound of $Ct^{-1/2}$ for each matrix game (t is the number of time the game is played). There are several algorithms with this regret rate, we refer the reader to [Han57, FS99]. In that case every element in (4.7) can be bounded by $\pi_t(s)R_t(s) = \frac{t_s(t)}{t} C_s (t_s(t))^{-1/2}$, where $t_s(t)$ is the amount of time spent in state s until time t. It follows that (4.7) can be bounded by:

$$\begin{aligned} \hat{r}_t - r^*(\pi_t) &\geq -\sum_s \frac{t_s(t)}{t} C_s t_s(t)^{-1/2} \\ &= -\frac{1}{t} \sum_s C_s t_s(t)^{1/2} \\ &\geq -\frac{1}{t} (\sum_s C_s^2 \cdot \sum_s t_s(t))^{1/2} \\ &> -\frac{1}{t^{1/2}} \sqrt{\sum_s C_s^2}\,, \end{aligned} \tag{4.8}$$

where the third inequality follows from Caushy-Shwarz inequality, and the last inequality from the definition of $t_s(t)$. **Q.E.D.**

Remark 3: It is interesting to note that the algorithm presented in the proof will work even for an unknown game. In [FS99], an algorithm with regret of $Ct^{-1/2}$ for an unknown game was introduced. By using the algorithm of [FS99] as the basic algorithm for each matrix game an algorithm that satisfies $\liminf_{n\to\infty} \hat{r}_t - r^*(\pi_t) \to 0$ with a guaranteed rate of $Ct^{-1/2}$.

4.2 The k-th Order Bayes Envelope

In this subsection we define the k-th order extension of the Bayes envelope for repeated matrix games. We show that the problem of attaining this envelope can be naturally embedded in a related single controller stochastic game. Recall that a matrix game is defined by the 3-tuple $(\mathcal{A}, \mathcal{B}, r)$ where $\mathcal{A}$ is P1's action set, $\mathcal{B}$ P2's action set and $r : \mathcal{A} \times \mathcal{B} \to \mathbb{R}$ is P1's reward. The Bayes envelope of a matrix game is defined as:

$$\hat{r}_t \geq r^*(\pi_t) \stackrel{\Delta}{=} \max_a \sum_b \pi_t(b) r(a, b)\,, \tag{4.9}$$

where $\pi_t \in \Delta(\mathcal{B})$ is the empirical measure of P2's actions by time t. The k-th order Bayes envelope is defined by taking into account the relative frequencies of k order action histories. To motivate the definition below, assume that P2 is stationary and that the conditional probabilities $p^k(b_t | b_{t-1}, \ldots b_{t-k})$ are known by P1 for any k-sequence $c_t = (b_{t-1}, \ldots b_{t-k})$ of P2's actions. Then

the average payoff that can be secured by P1 is

$$r^{*k} \triangleq \sum_{c\in\mathcal{B}^k} \pi^k(c) \max_a \sum_{b\in\mathcal{B}} p^k(b|c) r(a,b) = \sum_{c\in\mathcal{B}^k} \max_a \sum_{b\in\mathcal{B}} \pi^{k+1}(cb) r(a,b)\,,$$

where $\pi^k(c)$ is the probability of the k-sequence c of P2's actions, and cb denotes the concatenation of c and b. Accordingly, the k-order Bayes Envelope is defined as

$$\hat{r}_t \geq r^{*k}(\pi_t) \triangleq \sum_{c\in\mathcal{B}^k} \max_a \sum_{b\in\mathcal{B}} \pi_t^{k+1}(cb) r(a,b)\,, \tag{4.10}$$

where π_t^{k+1} is the empirical frequency of $k+1$ length sequences of P2's actions. Obviously, the standard Bayes envelope (4.9) is the 0 order Bayes envelope according to this definition.

An interesting property is that $r^{*k}(\pi_t)$ is monotone in k for a given empirical frequency, as can be easily verified. Thus, the k-th order Bayes envelope provides an improved payoff over envelopes of lower order.

It is well known that the Bayes envelope of a matrix game can be attained almost surely ([Han57]). The following theorem states that the k-th order Bayes envelope can be attained for any k.

Theorem 4.4 *For any given $k > 0$ there exists a strategy that attains the k-th order Bayes envelope (4.10) almost surely.*

Proof: The problem of attaining the k-th order Bayes envelope can be embedded in the following stochastic game, defined by:

1. $\mathcal{S} = \mathcal{B}^k$. Each state represents the last k actions of P2.
2. The action set of P1 is $\mathcal{A}$ in all states.
3. The action set of P2 is $\mathcal{B}$ in all states.
4. The state transitions are deterministic. The transition from state $s = bc \in \mathcal{B} \times \mathcal{B}^{k-1}$ is to the state $s' = cb'$, where b' is P2's action at state s. All other transitions have probability 0.
5. The reward function is $r(s,a,b) = r(a,b)$.

This stochastic game obviously is not recurrent, as P2 can preclude any state. However, since this is a single controller game, Theorem 4.3 implies that BE as defined in (4.3) is attainable. A simple interpretation of that envelope in terms of the original matrix game now yields the desired result. **Q.E.D.**

Remark 4: The rate of convergence to the k-th order Bayes envelope for a fixed k is $\Theta(t^{-1/2})$. The constant of the rate, C, is exponential in k as can be seen by reviewing (4.8), since the same game is played is every state, $C_s = C'$ for all s. This implies that $C = \Theta(B^k C')$, where B is the number of actions for P2. It should further be noted that a scheme that attains the k-order Bayes envelope *simultaneously* for all k may be devised, by slowly increasing k at an appropriate rate over time. Such a scheme can be termed "universal" as it attains the limit Bayes envelope.

4.3 Applications of Single Controller Games

In this sub-section we review two examples of classical problems in the realms of information theory and learning. We show that these examples fit well into our single controller schemes. The first example concerns prediction with expert advice (e.g. [Vov98]). The second problem concerns prediction in a finite channel, compared to a Markov predictor of a certain length. In the prediction with expert advice problem a single decision maker is repeatedly predicting the next outcome of an unknown channel. The decision maker is given advice from finitely many expert before each prediction. The decision maker's goal is to have his long term average performance as good as the best expert. This problem can be formulated as a dynamic game which satisfies our assumptions and a regret minimizing strategy is shown to exist. We make no assumptions regarding the experts and the channel. They may collaborate and for convenience are unified as a single player.

Consider a game of predicting the next letter in a channel with a finite alphabet of length A. Assume there are k experts $E^1, \ldots, E^k$. The game between the decision maker (P1) and the experts and channel (P2) is played in two phases for each time epoch n. At the first phase the experts state their predictions $E_n^1, \ldots E_n^k$, and then, knowing their predictions P1 makes his own prediction a_n and simultaneously (i.e. not knowing P1's prediction), P2 chooses the actual letter b_n. The reward obtained by P1 is $r(a_n, b_n) \in \mathbb{R}$. Although usually in prediction problems with finite alphabet, the reward is assumed a zero-one prediction reward, we do not restrict the model by this assumption.

Proposition 4.5 *There exists a strategy that guarantees a reward rate not lower than that of the best expert, almost surely.*

Proof: We actually prove a somewhat stronger result, namely that the reward rate is not lower than the reward for any stationary strategy that relies on the experts' joint advice. Obviously, each single expert is a special case. The prediction with expert advice is embedded in the following stochastic game that is defined by:

1. $\mathcal{S} = \{0, 1, \ldots, A^k\}$. State 0 is the first phase state and the other A^k states are the second

phase states - one for every possible vector of the experts' advice.

2. The actions of P1 in state 0 are $\mathcal{A}(0) = \{0\}$, that is no choice in state 0. In the second phase, P1 predicts that channel letter, so for $1 \leq s \leq A^k$, P1's actions are $\mathcal{A}(s) = \{1, \ldots, A\}$ which correspond to his prediction.

3. The actions of P2 in state 0 are all possible joint expert opinions, that is $\mathcal{B}(0) = \{1, 2, \ldots, A^k\}$. The possible actions of P2 at states $1 \leq s \leq A^k$, are $\mathcal{A}(i) = \{1, \ldots, A\}$ which corresponds to the actual letter to be predicted.

4. The transition probabilities are in $\{0, 1\}$. When moving from state 0 to each $1 \leq i \leq A^k$, the transition is deterministic and copies P2's action. From $1 \leq i \leq A^k$ back to 0 the state transition is with probability 1.

5. The reward at the first phase is 0, i.e. $r(0, \cdot, \cdot) = 0$ and the reward at the second stage is $r(s, a, b) = r(a, b)$, where $r(a, b)$ is the prediction cost.

This stochastic game obviously has a constant cycle time $\tau = 2$ and state 0 is the reference state. We conclude that all strategies are stable. As a result of Corollary 4.2 the set BE is attainable. That is:

$$\liminf_{n \to \infty} (\hat{r}_t - r^*(\pi_t)) \geq 0 \quad \text{a.s.}, \tag{4.11}$$

which holds for every letters sequence and expert advice. The strategies that participate in the maximum for $r^*(\pi_t)$, as in (4.2) certainly include the k strategies that always agree with a single expert i, $1 \leq i \leq k$. **Q.E.D.**

Remark 5: The proof implies that the average reward is in fact not lower than that of any stationary decision rule that relies on the experts' advice, in particular any weighted majority procedure.

The above result is well known for the 0-1 prediction loss. Our result slightly generalizes it by considering a general reward structure. A noted feature of the proof method is the ability to generalize to other games with similar structure, in which P2 (experts and channel) determine a state of nature. For example, one may consider a case in which if a certain letter appears in the channel then the stakes of the next stage game are doubled. One may also impose some restrictions (in the form of constrained Markov processes) on the dynamics of the channel and expert advice. These generalizations of the basic prediction with expert advice model easily fit our general scheme.

Let us now review the well known problem of universal prediction with a finite memory. Assume a game of prediction, similar to predicting the next letter above, this time the player tries to outperform all the possible finite memory predictors with memory span bounded by some

fixed N. At every time epoch t, P1 knows all the history of letters so far, $b_1, \ldots, b_{t-1} \in \mathcal{A}^{t-1}$, and makes his own prediction $a_t \in \mathcal{A}$. Simultaneously (i.e. not knowing P1's prediction), P2 chooses the actual letter b_t. The reward obtained by P1 is $r(a_n, b_n) \in \mathbb{R}$. A length N predictor is a mapping h from the last N letters to the next letter, $h : \mathcal{A}^N \to \mathcal{A}$. We are looking for a predictor that is as good as the best length N predictor, for every sequence of letters, asymptotically and almost surely. This classical information theory problem was solved by different methods for various situations, see [FM98] for a review. This problem is simply a special case of the k-th order Bayes envelope that was discussed in Subsection 4.2 with $\mathcal{B} = \mathcal{A}$ and typically $r(a, b) = 1_{a=b}$. Furthermore, as remark 4 states this result can be extended to prediction of any finite order.

It should be noted that the same method can be used for many other prediction games. As long as the reward is observed, more complex transition structure can be applied in practically the same manner. This is of particular interest for the prediction with finite memory, as some methods rely on this scheme and might be less trivial to extend to more complicated prediction situations.

5 A Game with Unattainable BE

In this section we analyze in detail a specific example of a stochastic game in which the empirical Bayes envelope is not convex and not attainable. This game is a single controller game in which P1 controls the state transitions. The game is also irreducible. The example sheds light on the structure of the Bayes envelope as a set and on some of its properties.

The game is defined by:

1. $\mathcal{S} = \{x, y\}$.
2. The actions of P1 in both states are $\mathcal{A} = \{0, 1\}$.
3. The actions of P2 in both states are $\mathcal{B} = \{0, 1\}$.
4. The transition probabilities are the same in both states and are determined by P1's action: $P(x|s, a = 0, b) = 0.99$, $P(y|s, a = 1, b) = 0.99$. Thus, P1 determines the next state by declaring it (up to a probability of 0.01).
5. The reward of P1 is determined by P2's action only : $r(s, a, 0) = 1$, $r(s, a, 1) = -1$.

The above as a zero sum game with average reward has a value of -1, as P2 will always play action 1. We study the case in which P2 does not necessarily play this minimax strategy because the game is either not zero-sum or alternatively P2 is not a rational player.

The space in which the Bayes envelope is defined is $\mathbb{R} \times \Pi_2$. We denote a point in this space by (r, π) where r is a real number and $\pi = (\pi(x,0), \pi(x,1), \pi(y,0), \pi(y,1))$ is the relative frequencies vector, where $\pi(s,b)$ is the relative frequency of state s and action b. The game is defined so that P1 determines the state frequencies and P2 determines the reward at each state. We now calculate the empirical Bayes envelope explicitly. The state frequencies are determined by P1's actions in both states. Assume P1 plays action 0 at state x with probability $p_{0|x}$ and plays action 0 at state y with probability $p_{0|y}$. Let $\pi(x)$ and $\pi(y)$ be the steady state probabilities of state x and y, respectively. The following relations hold:

$$\begin{aligned} \pi(x) &= 0.99\pi(x)p_{0|x} + 0.01\pi(x)(1-p_{0|x}) + 0.99\pi(y)p_{0|y} + 0.01\pi(y)(1-p_{0|y}) \\ \pi(x) &+ \pi(y) = 1 \end{aligned} \tag{5.1}$$

Which yields the solution:

$$\pi(x) = \frac{0.98p_{0|y} + 0.01}{1 + 0.98p_{0|y} - 0.98p_{0|x}} \tag{5.2}$$

Since $(p_{0|x}, p_{0|y}) \in [0,1] \times [0,1]$ are arbitrary it follows that P1 can set $\pi(x) \in [0.01, 0.99]$ at will. Recall from (3.3) that $r^*(\pi)$ is determined as the reward of the best reply strategy to the observed strategy of P2. The observed strategy $g = g(\pi)$ is defined by $g(b|s) = \frac{\pi(s,b)}{\pi(s)}$, where $\pi(s) = \sum_b \pi(s,b)$. Given g, if $g(0|x) - g(1|x) > g(0|y) - g(1|y)$ then P1 finds it more profitable to be in state x, while if the inequality is reversed then state y is preferred. As a result the best strategy of P1 is to have state x visited with frequency 0.99 if the above inequality holds, or have state y visited with frequency 0.99 if this inequality is reversed. If $g(0|x) - g(1|x) = g(0|y) - g(1|y)$ then any strategy of P1 will produce the same reward. As a result the Bayes envelope function is given by $r^*(\pi) = r^{BR}(g(\pi))$, where

$$r^{BR}(g) = \begin{cases} 0.99(g(0|x) - g(1|x)) + 0.01(g(0|y) - g(1|y)) & \text{if } g(0|x) > g(0|y) \\ 0.01(g(0|x) - g(1|x)) + 0.99(g(0|y) - g(1|y)) & \text{if } g(0|x) \leq g(0|y) \end{cases}$$

is the best response payoff of P1 to g.

We now proceed to show explicitly that the set $BE = \{(r, \pi) : r \geq r^*(\pi)\}$ is not attainable. For that purpose, it suffices to show that P2 can keep the average reward away from $r^*(\pi)$ for arbitrarily large times; more precisely, there exists $\epsilon > 0$ so that for any for any time T, there exists a strategy of P2 so that (with large probability) the average reward at some time beyond T will be ϵ-lower than the envelope. Given T, define the following strategy for P2:

$$g_t = \begin{cases} (1, 0, \frac{1}{2}, \frac{1}{2}) & 0 < t < T \\ (0, 1, \frac{1}{2}, \frac{1}{2}) & T \leq t < 2T \\ (0, 1, 1, 0) & 2T \leq t < 3T \end{cases}$$

Note that the entries of g_t correspond to $(g(0|x), g(1|x), g(0|y), g(1|y))$. A more detailed analysis can be found in C

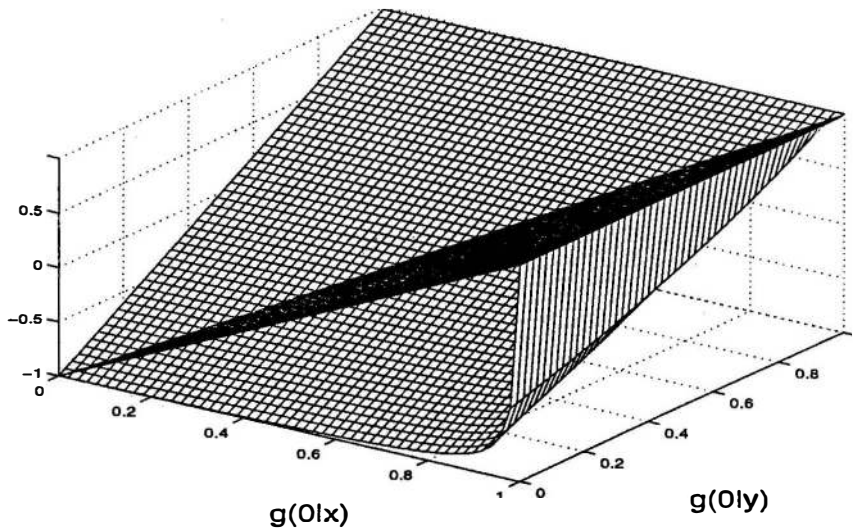


Figure 1: The upper envelope is BE , the lower envelope is induced by CBE .

Assume that T is sufficiently large so that the empirical frequencies are close enough to their expected value with probability close to 1. Note that for such T each state x or y is visited with frequency 0.01 at least. After the first T epochs, the observed strategy $g(\pi)$ equals $(1, 0, \frac{1}{2}, \frac{1}{2})$, and the best response payoff of P1 is 0.99. P1 is obliged to have the occupation frequency of state x near the maximum, or else the average payoff will be strictly less than $r^*(\pi)$. In the next T time epochs P1 should have the occupation frequency of state x near the maximum too. This is the case since otherwise the average reward at time $2T$ will be away from BE . Indeed, assuming for simplicity that the occupation frequency of state x by time T was exactly 0.99, an occupation measure of $0.99(1-\alpha)$ over the second T-period can be seen to yield an average reward (by $2T$) of $\hat{r}_{2T} = 0.99\alpha$, while the observed strategy is $g = (\frac{1}{2-\alpha}, \frac{1-\alpha}{2-\alpha}, \frac{1}{2}, \frac{1}{2})$ which gives a best response payoff of $r^*\pi_{2T} = 0.99\frac{\alpha}{2-\alpha}$. The two payoffs coincide only for $\alpha = 0$. Thus, P1 arrives at time $2T$ with average payoff (arbitrarily close to) 0, $\pi(x) = 0.99$, and $g(\pi) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Finally, at the last T-period, the Bayes reward will surely be denied, since the best-response reward will be determined by the observed strategy at state y, which (due to the relatively low occupancy of that state by $2T$) increases more quickly than possible for the average reward. Indeed, if P1 chooses $\pi(x) = \beta$ over that period ($0.01 \leq \beta \leq 0.99$), then we obtain an average reward of $\hat{r}_{3T} = \frac{1-2\beta}{3}$, while $g(\pi) = (\frac{0.33}{0.66+\beta}, \frac{0.33+\beta}{0.66+\beta}, \frac{1.01-\beta}{1.02-\beta}, \frac{0.01}{1.02-\beta})$ and consequently $r^*(\pi_{3T}) = 0.99\frac{1-\beta}{1.02-\beta} + 0.01\frac{-\beta}{0.66+\beta}$. A straightforward calculation shows that the difference between the average reward and the $r^*(\pi)$ remain bounded away from 0 (it is actually more than 0.33). Now the whole process can start all over again by taking a large enough interval T'. The same analysis applies, as the first $3T$ epochs are negligible. By repeating this construction BE is denied infinitely often.

In Figure 1, we show the performance envelopes for the game which is described above. The

planar coordinates are $g(0|x)$ and $g(0|y)$, the third dimension is the reward. In the figure, there are two surfaces. The upper surface is the envelope BE , the lower one is CBE . The CBE reward value is the guaranteed reward given that P2's strategy is characterized by $g(0|x)$ and $g(0|y)$. This value was calculated using linear programming for each point in the $g(0|x) \times g(0|y)$ plane. One can notice that when either $g(0|x)$ or $g(0|y)$ are extreme, the CBE is close to BE , however as they become central, CBE becomes less of a tight envelope, but still strictly above the value of the game which is -1.

6 Alternative Definitions of the Bayes Envelope

As opposed to repeated matrix games, where there is no ambiguity regarding how to define the Bayes envelope, in stochastic games a natural definition is more elusive. The main question is how to define an "average" strategy of the opponent based on the observed history of the game so far. As it turns out, a major complicating factor is the fact that the different states are visited with differing frequencies, which, moreover, depend on the choices of P1.

The approach presented so far assumes that the opponent's strategy is measured separately at each state, and correlations between the opponent's choices at different states are ignored. This approach allows to define in a simple way an average *stationary* policy of the opponent based on simple state-action frequency observations. However, to obtain an attainable envelope through this definition, an additional convexification of the resulting envelope is required. As a result, the guaranteed payoff may be lower than the best response to the observed stationary strategy of the opponent – that being the case even if the opponent is actually playing a stationary strategy.

In this section we examine two other possible definitions of the Bayes envelope, or equivalently of regret-minimizing strategies, for stochastic games. The first definition essentially reduces the game to a repeated super-game, with each repetition played over T time epochs (with T a large constant). Using techniques of repeated matrix games with incomplete observations, and letting the stationary deterministic strategies of P1 play the role of actions in this super game, one may strive to achieve the payoff that could have been achieved by the "best" of this strategies, in some appropriate sense. We shall formulate an appropriate Bayes-like envelope for this scheme, and argue that it enables to approximately obtain (with T large enough) the best-response to a stationary strategy of the opponent. A major drawback of this scheme, which effectively ignores the state dynamics of the game, is that it requires explicit experimentation of *all* stationary deterministic strategies, which leads to a poor convergence rate.

The second approach we consider replaces the constant intervals of the previous approach with renewal times to the reference state. This leads to a simpler definition of the "average strategy" for the opponent which averages the strategies observed over each renewal interval. However,

the resulting envelope turns out to be unattainable in general, and we show that this may be attributed specifically to uneven renewal cycle times.

6.1 Average Strategies Over Fixed Time Intervals

Let us partition the time horizon into blocks of length T. We will look at the stochastic game as a repeated matrix game, with each length-T play considered a single shot of this game. Over each time interval, P2 may apply a general (mixed, history dependent) strategy, which is to be considered as a mixed action in the super-game. Note that the super-game played over each time interval depends on the initial state at the beginning of that interval; it will be convenient to suppress this dependence by allowing P2 to choose the initial state as part of his strategy. Let P2's strategy in the k-th interval (including the choice of the initial state) be denoted by σ_k^2. The average strategy by time kT is then defined as $\hat{\sigma}_k^2 = \frac{1}{k}\sum_{i=1}^{k}\sigma_i^2$. The best response reward r^{BR} may now be defined as

$$r_k^{BR} \stackrel{\triangle}{=} \max_{f\in F} r(f, \hat{\sigma}_k^2)\,, \tag{6.1}$$

where F is the set of stationary strategies of P1.

We argue that the envelope defined by r^{BR} above may be attained asymptotically. Note that the strategies σ_k^2 are not in general observed by P1; however, the reward over each T-interval is known. Regret minimizing strategies for repeated games with partial observations have been considered in [FS99] and [Rus99], and may be applied in a straightforward manner to our repeated super-game. Indeed, observing that the maximum in (6.1) is achieved over the finite set of deterministic stationary strategies, these strategies may be taken as the finite action set for P1 in the super-game. Furthermore, given that this can be attained in the modified game where P2 can choose the initial state at will, it is certainly attained in the actual game where P2 is constrained in his choice by the state dynamics.

It is illuminating to evaluate the best-response function when P2 plays a stationary strategy g, with T large. In this case, the average strategy converges to g (restricted to T time steps), and for T large enough r^{BR} will be arbitrarily close to the best response of P1 to g in the original stochastic game. In fact, one may consider a modified scheme which slowly increases the interval length T to infinity; using an appropriate rate of increase, the best response of P1 to g may be precisely achieved in the long run.

The above approach suffers from certain disadvantages. First, it relies on explicit experimentation of all deterministic stationary strategies, and the consequent convergence rate will be prohibitive in all but the simplest problems. Furthermore, the dynamics of the system is completely ignored and no use is made of the state transition structure. In comparison, the convergence rate to the empirical Bayes envelope depends mainly on the reference-state renewal time.

Finally, the average strategy expression used in (6.1) is cumbersome due to the need to consider T-step strategies, rather than simpler (e.g. stationary) alternatives. A notable feature of the method just described is that it attains the best response reward if the opponent is stationary.

6.2 Average Strategies over Renewal Times

As a possible variant of the previous scheme, one may consider replacing the arbitrary T-length intervals with renewal intervals, between consecutive visits to the reference state. This approach may seem more natural for a repeated game reduction, since it inherently eliminates the problem of different initial states. Furthermore, it fits well with our renewal-based scheme for approachability in stochastic games as presented in Section 2.

The basic notion of a strategy for P2 here is the strategy per cycle (Note that this is an infinite class unless the cycle time is bounded). As in the previous scheme this strategy is not fully observed by P1; however both the reward per cycle and the cycle time are measured. A basic problem here is that the cycle time is not constant, and depends on the choices of P1. This leads to difficulties in defining an average an "average strategy" of P2, or of a feasible reward objective for P1. While several options exist for such a definition, none has been found to lead to an attainable and meaningful goal. A more detailed consideration of this issue is provided in Appendix B.

7 Conclusion

We have presented an approach to regret minimization in stochastic games, by considering a Bayes-like envelope in the state-action space. The basic object which underlies this approach is BE , the empirical Bayes envelope. which is based on extracting a stationary strategy of the opponent directly from the observed state-action frequencies.

While BE is not attainable in general, it was shown that its convexification leads to an attainable envelope, the CBE . This envelope provides some performance guarantees that qualify it as a goal for adaptive play. In particular, the guaranteed average payoff is never less than the value of the zero-sum game, and is strictly higher that the value if the other player deviates from an optimal (minimax) strategy, provided all states are visited with non-negligible frequencies. In the case of single-controller games, we have considered some applications where this approach is applied to specific problems, and easily leads to well known results with possible non-trivial generalizations.

Some open questions remain concerning the model of this paper. First, while approaching strategies for the proposed envelopes follow directly from the approachability framework, it would

be interesting to have such strategies formulated in a more direct way, perhaps in the style of [FS99]. Even more importantly, one may examine whether certain strategies that attain the CBE also provide improved performance against specific classes of strategies of the opponent, such as his stationary strategies. Second, if the empirical Bayes envelope is not approachable, what can be said on it – what are the conditions in which it is deniable, or deniable in some fixed time sense (as in [MS00b]). Third, the Pareto efficiency of the convex Bayes envelope or appropriate subsets thereof should be further studied. Fourth, additional applications of the theory should be found in areas that include communications over arbitrary channel, prediction, utilization of expert advice, competitive queuing problems ([SS93]), and others.

We have briefly considered in Section 6 some alternative definitions for the Bayes envelope. These definitions rely on the space of general strategies of the opponent, and lead to relatively complex schemes, with convergence rates that are exponential in the number of states. Possible simplified schemes of this nature are of major interest for further research.

Finally, the concepts of regret minimization may be applied for learning situation, where the system parameters are not known in advance, or the problem is too complex for explicit solution. Using a Reinforcement Learning approach (e.g. [BT95, KLM96]) may yield algorithms which are adaptive and in some sense optimal for a large class of game problems.

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A Appendix

In this Appendix we provide a generalized framework for approachability in stochastic games, and apply it to establish the attainability of CBE under this more general setup. The approachability results here extend those of [SS93] and are presented in detail in [MS00a] along with the proofs, we shall only state here the specific result that is required for our purpose.

The main assumption in [SS93], and in the body of the present paper, is the existence of some reference state which is recurrent under *all* stationary strategies. In principle, however, weaker requirements that result in a value that does not depend on the initial state can be employed to develop similar results. Specifically, we require here that P1 has *some* strategy that ensures a return to the reference state. We refer to this model as *0-communicating*. Note that to guarantee that the value is independent of the initial state, a similar requirement needs to be imposed on P2 as well. However, The one sided assumption suffices for the purpose of developing

sufficient approachability conditions for P1 (necessary and sufficient conditions and the approachability/excludability duality are expanded in [MS00a]). Note further that in a single player model (MDP) the 0-communicating requirement is weaker than the standard communicating class assumption.

To give an example where the weakened assumption may be useful, consider a queuing system with P1 the server and customer arrivals modeled as P2 (cf. [SS93]). A natural reference state in a stable system is the empty queue, however a possible strategy for P1 is to stop service and prevent the queue from emptying. The present framework clearly accounts for this case.

We note that the weaker assumptions may entail certain complications in the approaching strategies. An approaching strategy is composed of piecewise stationary strategies, each optimal in the designated projected direction. In the absence of a reference state which is necessarily reached under those optimal strategies of P1, one may need to play each strategy over unboundedly increasing periods of times to ensure that the average payoff over each interval is arbitrarily close to the optimal one.

The basic assumption in this Appendix is the following:

Assumption 2 *There exists a state* $0 \in \mathcal{S}$ *and a stationary (possibly randomized) strategy* f^* *of P1 such that state 0 is reached with probability 1 for all strategies of P2 and from all initial states* $\forall s \in \mathcal{S}$.

Note that the apparently weaker assumption that allows f^* to depend on the initial state is equivalent to the above. The existence of such a strategy can be checked by inspecting the transition structure. Further note that due to the finiteness of the state space, the existence of a uniform bound on $E(\tau^2)$ over all initial states and strategies of P2 is guaranteed, where τ is the time to state 0, which is a basic requirement for an approachability theorem.

We can now define in an analogue manner to Section 2 the sets $M(F,g)$. As the expectation from different states is no longer the same we refer to the expectation starting from state 0. $M(F,g) \stackrel{\Delta}{=} \text{co}(\{\lim_{n\to\infty} E^0_{f,g}\hat{rn}_n, f \in F\})$ The following theorem is a part of a general characterization of convex approachable sets given in [MS00a].

Theorem A-1 *Let Assumption 2 hold. Let* B *be a closed convex set in* $\mathbb{R}^k$. *Then* B *is approachable if* $M(F,g) \cap B \neq \emptyset$ *for every stationary strategy* $g \in G$.

Returning to CBE and BE , the same argumentation as in Section 3 can be applied so that BE is defined for games that satisfy the weaker Assumption 2. The reward in question $r_0(f,g) \stackrel{\Delta}{=} \lim_{n\to\infty} E^0_{f,g}(\hat{r}_n)$ is the average reward from initial state 0 under the strategy pair f and g.

Corollary A-2 *Suppose Assumption 2 holds. Define BE as:*

$$BE \triangleq \{(r,\pi) : r \geq r^*(\pi) \triangleq \max_{f\in F} \min_{g\in G(\pi)} r_0(f, g(\pi))\}. \tag{A-1}$$

Then $CBE = co(BE)$ is approachable and the resulting envelope attainable (from any initial state).

Proof: The result follows in the same manner as Theorem 3.2 since Assumption 2 applies and Theorem A-1 can be applied. Approachability implies attainability since $r^c(\pi)$ is continuous. This can be shown because $r^*(\pi)$ is lower-semi-continuous as in the proof of Proposition 3.1. The lower-semi-continuity follows because $E^0_{f,g}$ is continuous in g for any given f (e.g. [FV96], Proposition M.6 and noting that the probability of arriving to each irreducible set is continuous is the transition probabilities which are continuous in f and g.). **Q.E.D.**

B Appendix

This appendix is dedicated for a more detailed analysis of the example for an unattainable empirical Bayes envelope that was reviewed in Section 5. Recall that

$$g(\pi) = \left(\frac{\pi(x,0)}{\pi(x)}, \frac{\pi(x,1)}{\pi(x)}, \frac{\pi(y,0)}{\pi(y)}, \frac{\pi(y,1)}{\pi(y)}\right)$$

So that the best response reward is (5.3) (in π):

$$r^*(\pi) = \begin{cases} 0.99\frac{\pi(x,0)-\pi(x,1)}{\pi(x)} + 0.01\frac{\pi(y,0)-\pi(y,1)}{\pi(y)} & \text{if } \frac{\pi(x,0)-\pi(x,1)}{\pi(x)} > \frac{\pi(y,0)-\pi(y,1)}{\pi(y)} \\ 0.01\frac{\pi(x,0)-\pi(x,1)}{\pi(x)} + 0.99\frac{\pi(y,0)-\pi(y,1)}{\pi(y)} & \text{if } \frac{\pi(x,0)-\pi(x,1)}{\pi(x)} \leq \frac{\pi(y,0)-\pi(y,1)}{\pi(y)} \end{cases}$$

We now proceed to calculate explicitly that the set $BE = \{(r,\pi) : r \geq r^*(\pi)\}$ is not attainable. Recall P2's strategy:

$$g_t = \begin{cases} (1,0,\frac{1}{2},\frac{1}{2}) & 0 < t < T \\ (0,1,\frac{1}{2},\frac{1}{2}) & T \leq t < 2T \\ (0,1,1,0) & 2T \leq t < 3T \end{cases}$$

The relative frequencies at time T by the above choice satisfy:

$$\pi_T = (\theta, 0, \frac{1-\theta}{2}, \frac{1-\theta}{2}) + \epsilon \tag{B-1}$$

where $\theta \in [0.01, 0.99]$ and $\epsilon \in \mathbb{R}^4$ is a small vector ($||\epsilon||_\infty << 0.01$). P1 may choose θ as he wishes, however $r^*(\pi_T) = 0.99$ and the average reward is $\hat{r}_T = \theta + \eta$ (η will be used as a generic small positive scalar). We therefore conclude that unless $\theta = 0.99$ BE is denied at time T.

During the second T epochs, the reward is -1 for state 0 and 0 on the average for the second state. We therefore have that:

$$= \frac{1}{2}\pi_T + \frac{1}{2}(0, \gamma, \frac{1-\gamma}{2}, \frac{1-\gamma}{2}) + \epsilon \quad \text{(B-2)}$$

$$= (0.495, \frac{\gamma}{2}, \frac{1}{400} + \frac{1-\gamma}{4}, \frac{1}{400} + \frac{1-\gamma}{4}) + \epsilon \quad \text{(B-3)}$$

where $\gamma \in [0.01, 0.99]$, as before is determined by P1 and ϵ a small noise. The average reward at time $2T$ is $\hat{r}_{2T} = 0.495 - \frac{\gamma}{2} + \eta$. The Bayes reward $r^*(\pi_{2T})$ satisfies:

$$r^*(\pi_{2T}) \geq 0.99\frac{0.495 - \frac{\gamma}{2}}{0.495 + \frac{\gamma}{2}} + \eta \quad \text{(B-4)}$$

It can be verified by simple algebra that if $r^*(\pi_{2T}) - \hat{r}_{2T}$ is small, then $\gamma = 0.99$. In the last T time epochs, P2 plays so $g(x) = 0$ and $g(y) = 1$. π_{3T} is therefore:

$$\pi_{3T} = \frac{2}{3}\pi_{2T} + \frac{1}{3}(0, \delta, 1-\delta, 0) + \epsilon \quad \text{(B-5)}$$

$$= (0.33, 0.33 + \frac{\delta}{3}, \frac{1}{300} + \frac{1-\delta}{3}, \frac{1}{300}) + \epsilon \quad \text{(B-6)}$$

where $\delta \in [0.01, 0.99]$ and ϵ is as before. The average reward is $\hat{r}_{3T} = (\frac{1-2\delta}{3}) + \eta$ while the Bayes reward is obtained by spending more time in state y and is:

$$r^*(\pi_{3T}) \geq 0.99\frac{1}{3}\frac{1-\delta}{\frac{1}{150} + (\frac{1-\delta}{3})} - 0.01\frac{\frac{\delta}{3}}{0.66 + \frac{\delta}{3}} + \eta \quad \text{(B-7)}$$

A straightforward optimization shows that in that case $r^*(\pi_{3T}) - \hat{r}_{3T} > 0.33$. This implies that for every small enough $\eta > 0$ there exists some time by which $r^*(\pi^t) - \hat{r}^t > \eta$ for some $T \leq t < 3T$ or that at $3T$ $r^*(\pi_{3T}) - \hat{r}_{3T} > 0.33 > \eta$. This shows that either BE is excluded in times T or $2T$ or $3T$.

C Appendix

In this appendix we explore possible notions of optimality in the strategy space, as per our discussion in Section 6. The main difficulty is how to define an average strategy. We propose two different definitions of the average strategy and provide examples that show that the Bayes envelope in the strategy space may be non attainable. One way to define an average strategy is to measures average strategy for each time epoch, a different way is to measures the average strategy only at the renewal times. Assume that at every renewal time P2 decides what strategy to play until the next renewal time. This decision is private and P1 observes the parts of P2's strategy that become relevant as the game evolves. For repeated matrix games with partial observation it was shown in [Rus99] that this approach results in an attainable envelope which equals the Bayes

envelope. The critical issue is the fact that the cycle time is an important factor so that strategies that induce slow cycle time make it hard for the average reward to change fast so it may become lower than the Bayes reward.

Recall that the space of strategies of P2 is denoted by Σ_2. Suppose that by time t the average strategy is defined by an empirical measure on Σ_2, $\psi \in \Delta(\Sigma_2)$. The Bayes envelope with respect to the strategy space is defined by:

$$r^s(\psi) = \sup_{f \in F} \frac{\int \rho(f, \sigma_2) d\psi}{\int \tau(f, \sigma_2) d\psi}, \tag{C-1}$$

where $\rho(f, \sigma)$ and $\tau(f, \sigma)$ are the average reward per cycle and the average time per cycle, when P1 plays f and P2 plays σ, respectively.

As mentioned before, the measurement of the average strategy is problematic. First, suppose that the average strategy is the average strategy per time epoch. This means that is strategy ζ_1 was played for τ_1 epoch and strategy ζ_2 was played for τ_2 then the average strategy is $\hat{\sigma_2} = \frac{\zeta_1 \tau_1 + \zeta_2 \tau_2}{\tau_1 + \tau_2}$, and generally, if strategies $\zeta_1, \ldots, \zeta_k$ were played for intervals of $\tau_1, \ldots, \tau_k$ then the average strategy is:

$$\hat{\sigma_2} = \frac{\sum_{i=1}^k \zeta_i \tau_i}{\sum_{i=1}^k \tau_i}. \tag{C-2}$$

We now give an example in which r^s is not attained even though P2 plays a stationary strategy. The difficulty is the result of the fact that P1 observes a different strategy than the one actually played. This is so since slow cycles weigh more than fast cycles. Let us study the following game, the game is written as a matrix game, however for every entry i, j there are two numbers - the reward per cycle r_{ij} and the cycle time τ_{ij}. The meaning is that if P1 picked i and P2 picked j then P1 obtains a total reward of r_{ij} over the entire cycle time τ_{ij}. One can embed this game in a stochastic game by assuming that the actions for P1 and for P2 are taken only in the reference state and after these actions take place, the game is deterministic until the next return to the reference state which occur after τ_{ij} time epochs and the reward during the cycle is r_{ij}. P2's strategy between the returns to the reference state is defined by P2's action in the reference state. Let us denote the relative frequencies of the actions in this state by $\alpha \in \Delta^J$. The optimal average reward against a given strategy g is:

$$r^s(\alpha) = \max_i \frac{\sum_j r_{ij} \alpha(j)}{\sum_j \tau_{ij} \alpha(j)} \tag{C-3}$$

For our purposes it suffices to examine a case of a 2×2 matrix. In that case, let α be the frequency of action 1 at the reference state. Let the game in interest be defined by:

$$G = \begin{pmatrix} (70, 1) & (-20, 6) \\ (-30, 4) & (1, 1) \end{pmatrix}.$$

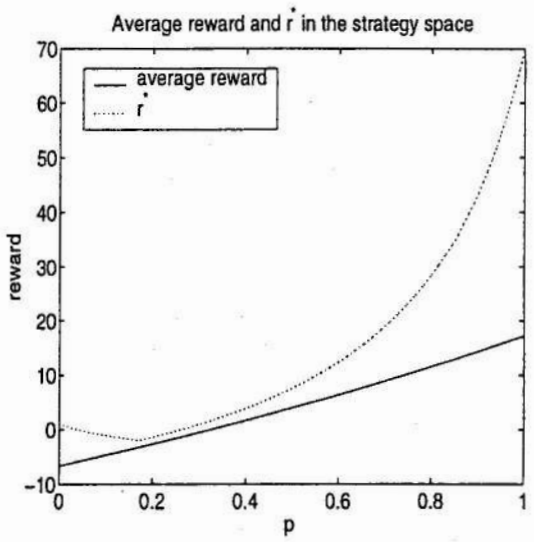


Figure 2: $r^*(p)$ and $\hat{r}(p)$ when measuring strategy on a time base, note that $r^*(p)$ is strictly above $\hat{r}(p)$

Suppose that P2's strategy is to play action 1 with probability $\alpha = 0.7$. We allow P1 to play whatever he wishes, that is to choose action 1 (row action) with probability p at every renewal time. The average reward is:

$$\hat{r}(p) = \frac{p\alpha r_{11} + p(1-\alpha)r_{12} + (1-p)(\alpha)r_{21} + (1-p)(1-\alpha)r_{22}}{\tau_1(p) + \tau_2(p)}$$

where

$$\tau_1(p) = (p\alpha\tau_{11} + p(1-\alpha)\tau_{21})$$
$$\tau_2(p) = (p\alpha\tau_{12} + p(1-\alpha)\tau_{22})$$

The problem with this approach is that P1's strategy, as reflected by p, affects the observed strategy of P2. The observed average strategy of P2 is:

$$\alpha'(p) = \frac{\tau_1(p)}{\tau_1(p) + \tau_2(p)}.$$

The optimal reward against $\alpha'(p)$ is given by (C-3). Suppose P2 plays $\alpha = 0.7$ for a long time (long enough that everything holds with a high probability). P1 may choose whatever p he likes, as a result the measured strategy of P2 is $\alpha'(p)$ and the optimal reward $r^s(\alpha'(p))$. The average reward $\hat{r}(p)$ and the optimal reward against the measured strategy $r^s(\alpha'(p))$ are given in Figure 2. As it turns out, for this game $r^s(p) > \hat{r}(p) + \frac{1}{2}$ for all p. This implies that the Bayes reward in the strategy space is denied.

We now suggest a different method to measure the average strategy. Suppose that at every time the reference state is visited P2 decides which strategy to play until the next visit to the

reference state. Since we saw that adding the varying cycle time factor into the measurement of the average strategy results in an unattainable goal, even against stationary strategies, let us eliminate this factor. For this end let the average strategy be defined by:

$$\hat{\sigma_k} = \frac{1}{k}\sum_{i=1}^{k}\sigma_i\,,$$

where σ_i is as before. The optimal reward is given as in (C-1). Let us give an example in which the resulting envelope is not attainable. Note that as opposed to the previous case where a stationary strategy by P2 was sufficient, this is not the case anymore. If P2 plays a stationary strategy then P1 can guarantee attaining the envelope by playing the optimal (pure) action against P2's stationary strategy, P1's measurements are not "biased" so that the envelope may be attained.

The example is written in a similar manner to the previous example, let g be the frequency of action 2 at the reference state. Let the game in interest be defined by:

$$G = \begin{pmatrix} (2,1) & (1,2) \\ (3,2) & (3/25, 3/5) \end{pmatrix}.$$

A straight-forward calculation yields that:

$$r^s(g) = \max\left\{\frac{2-g}{1+g},\ 3-\tfrac{144}{50}g,\ 2-\tfrac{7}{5}g\right\} \qquad \text{(C-4)}$$

The function $r^s(g)$ is presented in Figure 3. Note that it is not convex and that the regions in which the action $i = 1$ is dominant are not connected. Let us define the following strategy of P2:

$$g(n) = \begin{cases} 0.2 & 0 < n \leq N \\ 0.9 & N < n < 2N \end{cases}.$$

N is an arbitrary chosen large number so that the following example can be considered to hold with high probability. Note that the strategy is defined with relation to the renewal epochs and not with relation to time. This means that a mixed action of $g = 0.2$ will be taken exactly N times. For the strategy $g = 0.2$, action 1 is better. Since P1 tries to attain $r^s(g)$ he should play action 1 most of the first N decisions. A straight-forward calculation shows that the average reward at the $2N$ return to the reference state is $\hat{r}_{2N}(p) = \frac{2.9-0.692p}{3.1-1.16p}$, where p is the probability that action 2 was chosen by P1 at the second N decisions in the reference state. At the $2N$ return to the reference state the average reward $\hat{r}_{2N}$ satisfies that $\hat{r}_{2N}(p) < 1.13$, however the average strategy of P2 is with $g = 0.55$. For $g = 0.55$ the optimal action is action 2 and the reward that would have been expected is 1.15. This implies that $r^s(g)$ is denied either at the N or at the $2N$ return to the reference state . The game can now be played all over again with N' exponentially larger than N so that the first $2N$ returns are not significant and thus $r^s(g)$ can be denied infinitely often.

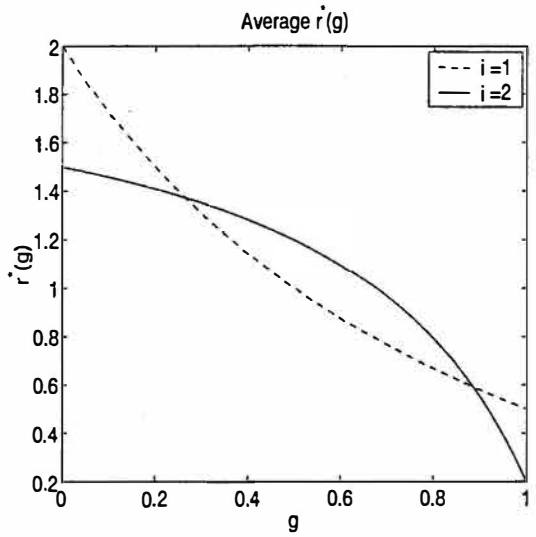


Figure 3: The two graphs contribute to $r^*(g)$ which is the maximum of both of them

There is an interesting intuition behind the above example. Examine the three dimensional space that is defined by the action frequency g, the average reward per cycle $\hat{r}$ and the average time per cycle $\hat{\tau}$. The envelope that is to be attained in that space is defined as:

$$\left\{(\hat{r},\hat{\tau},g) : \frac{\hat{r}}{\hat{\tau}} \geq r^*(g)\right\} . \tag{C-5}$$

This envelope turns out to be not connected, since the action $i = 1$ has two regions in which it is optimal. The lack of connectivity is crucial since by switching from two areas in which $i = 1$ is optimal the convex hull is not in the envelope. The above example demonstrates the fundamental difference of the shape of the Bayes envelope between stochastic and matrix games. While in matrix games $\tau = 1$ the envelope (C-5) is convex and certainly connected this is not the case for stochastic games. The lack of connectivity and convexity prevent the attainability of the BE in the strategy space.

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